

Bishal Das

Machine Learning Engineer | Data Engineer | AI Developer

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Professional Summary

Computer Science student focused on Machine Learning, Data Engineering, and Computer Vision. Skilled in Python, TensorFlow, PySpark, and Kafka. Built ML models, real-time data pipelines, and automation tools, IOT.

Technical Skills

Languages: Python, SQL, C++, Java, Kotlin, Bash

ML/AI: TensorFlow, PyTorch, Scikit-learn, Keras, XGBoost, OpenCV, MediaPipe,

Data Engineering: Apache Spark, PySpark, Apache Kafka, ETL, Stream Processing

Data Science: Pandas, NumPy, Feature Engineering, EDA, Statistical Analysis, Matplotlib, Seaborn, Plotly

Tools & Databases: Git, Docker, MySQL, PostgreSQL, Streamlit

Robotics: ROS 2, Gazebo, URDF, Gesture Recognition

Education

B.Tech in Computer Science and Engineering | ICFAI University, Tripura

2024 – 2027 (Expected)

CGPA: 9.3

Diploma in Mechanical Engineering | Tripura Institute of Technology

2021 – 2024

CGPA: 7.5

Projects

Apache Spark Real-Time Data Pipeline | *PySpark, Kafka, SQL*

- Processed **20,000+** records and improved query speed by **40%** using partitioning and caching.
- Built Kafka-based streaming pipeline with **sub-second** ingestion latency.

AIRA 2.0 – AI Voice Assistant | *Python, NLP, TF-IDF*

- Achieved **90%** intent accuracy across **500+** commands.
- Added **15+** automation features for file, web, and app control.

Phish-Check – Malicious URL Detector | *Python, Random Forest, Streamlit*

- Trained model on **10,000+** URLs with **30+** features; achieved **91%** accuracy.

QUADBOT – Gesture-Controlled Robot | *ROS 2, MediaPipe, OpenCV*

- Developed gesture-controlled robot with command latency below **100 ms**.

Smart Dustbin – Edge AI Waste Classifier | *Raspberry Pi, CNN, TensorFlow Lite*

- Deployed CNN model with **85%** accuracy and inference below **200 ms**.

Work Experience

AI Research Intern | National Institute of Technology (NIT) Silchar

Jun 2025 – Jul 2025

Hybrid | Silchar, Assam, India

- Built ML-based smart irrigation system; reduced water use by **30%** with **88%** prediction accuracy.
- Integrated Firebase IoT backend for real-time sensor sync with sub-second latency.
- Developed pipeline for automated irrigation decisions from live sensor data.

Machine Learning

Jan 2024 – Present

- Developed and deployed **5+** Streamlit ML applications for practical use cases.
- Built Smart Dustbin solution for real-time waste classification on Raspberry Pi.

Certifications

AI/ML Internship Certificate – National Institute of Technology (NIT) Silchar

Jul 2025

Achievements

- Built a **Smart Dustbin** using Raspberry Pi and CNN for real-time waste segregation .
- Presented projects at multiple **technical events**, demonstrating practical AI and robotics applications.
- Developed a **Hexapod robot** and showcased it at **Startup Tripura**, secured **Ranked among the Top 5 projects** ,and awarded funding support.